

Section 13

Midterm Review

Topics: everything you've learned up to this point!

Section 12

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Midterm Study Guide

These questions are here to help you review and practice some potential problems. There's no guarantee the questions on these slides will be similar to questions on the midterm. There are also a variety of topics we have covered that is not covered in this study guide.

Make sure to review slides, homeworks, textbooks and labs. Good luck! :)

Pointers

Pointers are an area that a lot of people struggle with, don't worry, they're really simple!

Computers deal with memory, everything you do is memory related. When you execute a program, when you create a variable, and even when you load a file, it gets loaded into memory. There is nothing you can do without memory.

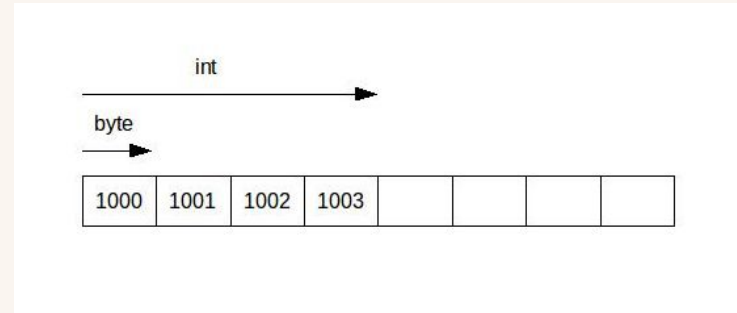
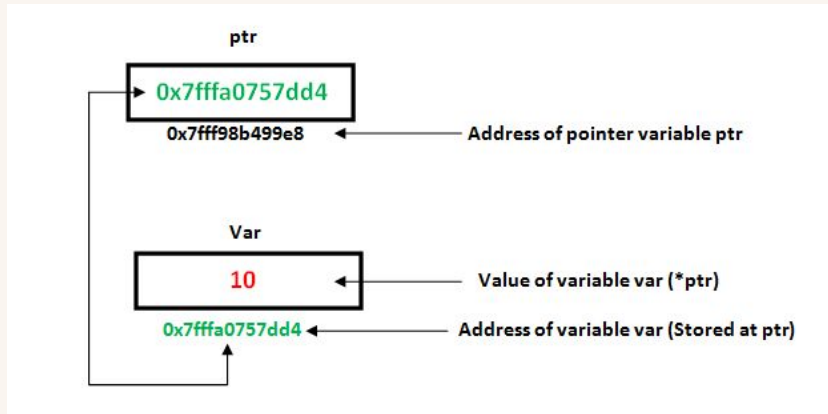
A pointer is an *integer* that stores a *memory address*

Forget types, they were something we made up to make our lives easy, they aren't real!

Pointers

Imagine a really long road with houses along them, each of the houses have an address. That's what memory is, a linear 1D line. We need a way to address it!

We need a way to read/write from these specific addresses, and that's what pointers let us do.



Pointers

```
int x = 7;  
int* ptr_to_x = &x;  
// int* holds the memory address of a (presumed) int  
// &<var> gets the memory address of the var  
int y = *ptr_to_x;  
// *<var> gets the value inside a pointer
```

What is the output?

What is the output of the following code:

```
int b = 6;  
int* a = &b;  
b += 2;  
cout << *a << endl;
```

```
int x = 4;  
int y = 6;  
  
int* a = &x;  
int* b = &y;  
  
cout << (*a) * (*b);
```

Short Questions

Do the following math computations in C++:

- a. $3 * 1 / 2$
- b. $3.0 / (5/2)$
- c. $2 * 'a'$
- d. $1 + 3 / 2 * 2$

What does this code do (in english)?

```
int function(int num){  
    int lastnums = num % 100;  
    return lastnums;  
}
```

Short Questions

Do the following math computations in C++:

- | | |
|--------------------|-----|
| a. $3 * 1 / 2$ | 1 |
| b. $3.0 / (5/2)$ | 1.5 |
| c. $2 * 'a'$ | 194 |
| d. $1 + 3 / 2 * 2$ | 3 |

What does this code do (in english)?

```
int function(int num){  
    int lastnums = num % 100;  
    return lastnums;  
}
```

This function gives me the last 2 digits of any number

Short Questions

What are the outputs of the following array operations

- a. `{‘a’, ‘b’, ‘c’}[1] = ?`
- b. `{-5, 6, 2, 8}[3] = ?`
- c. `“euler”[2] = ?`
- d. `{“math”, “compsci”, “art”}[1][4]`

What header would you write for the following functions:

- a. ``increment`` takes in an integer ref, and doesn’t return a value
- b. ``countE`` takes in a string, and returns an integer
- c. ``swap`` takes in two double references, and doesn’t return a value

Short Questions

What are the outputs of the following array operations

- a. `{'a', 'b', 'c'}[1] = 'b'`
- b. `{-5, 6, 2, 8}[3] = 8`
- c. `"euler"[2] = 'l'`
- d. `{"math", "compsci", "art"}[1][4] = 's'`

What header would you write for the following functions:

- a. ``increment`` takes in an integer reference called ``a``, and doesn't return a value
 - i. `void increment(int& a);`
- b. ``countE`` takes in a string, and returns an integer
 - i. `int countE(string s);`
- c. ``swap`` takes in two double references, and doesn't return a value
 - i. `void swap(double& a, double& b);`

Short Questions

What is the output of the following string operations:

- a. `“sadab”.substr(2,3);`
- b. `“yoomin”.substr(3);`
- c. `“rawad”.substr(0,3);`
- d. `“markus”.substr(1,3);`

Write a C++ conditional to represent the following cases:

- a. `[60, 61, 62, ..., 80]`
- b. `[-∞, ... -12, -11, 11, 12, ..., ∞]`
- c. `[-∞, ..., -4, -2, 0, 2, 4, ..., ∞]`
- d. `['c', 'd', 'e', ..., 'w', 'x']`

Short Questions

What is the output of the following string operations:

- | | |
|--------------------------|-----|
| a. “sadab”.substr(2,3); | dab |
| b. “yoomin”.substr(3); | min |
| c. “rawad”.substr(0,3); | raw |
| d. “markus”.substr(1,3); | ark |

Write a C++ conditional to represent the following cases:

- | | |
|---|----------------------------------|
| a. [60, 61, 62, ..., 80] | $i \geq 60 \ \&\& \ i \leq 80$ |
| b. $[-\infty, \dots -12, -11, 11, 12, \dots, \infty]$ | $i > 10 \ \ i < -10$ |
| c. $[-\infty, \dots, -4, -2, 0, 2, 4, \dots, \infty]$ | $i \% 2 == 0$ |
| d. ['c', 'd', 'e', ..., 'w', 'x'] | $c \geq 'c' \ \&\& \ c \leq 'x'$ |

Short Questions

What is the output of the following random operations:

- a. `rand();`
- b. `rand() % 100;`
- c. `rand() % 10 + 1;`
- d. `rand() % 100 + 900;`

Give the terminal instructions to do the following to `file.cpp`

- a. Compile the file to the default executable
- b. Compile the file to an executable named ``my_executable``
- c. Run the file called ``my_executable``
- d. Run the file called ``my_executable`` with input redirection from ``my_data.csv``

Short Questions

What is the output of the following random operations:

- | | |
|-------------------------------------|--------------------------|
| a. <code>rand();</code> | <code>[0,INT_MAX]</code> |
| b. <code>rand() % 100;</code> | <code>[0, 99]</code> |
| c. <code>rand() % 10 + 1;</code> | <code>[1,10]</code> |
| d. <code>rand() % 100 + 900;</code> | <code>[900, 999]</code> |

Give the terminal instructions to do the following to file.cpp

- Compile the file to the default executable
- Compile the file to an executable named ``my_executable``
- Run the file called ``my_executable``
- Run the file called ``my_executable`` with input redirection from ``my_data.csv``

What is the output?

```
#include <iostream>
using namespace std;

int main(){
    int product = 1;
    for(int i = 0; i>=-3; i-=4){
        product *= i;
    }
    cout << product << endl;
}
```

```
#include <iostream>
using namespace std;

int main(){
    int count = 0;
    for(int i = 3; i>=-1; i-=2){
        count++;
    }
    cout << count << endl;
}
```

Implementation

Write code that performs the quadratic formula and stores the values of the roots.

It takes in 5 values:

- 2 references to doubles (x, y)
- 3 double values (a, b, c)

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Implementation

Write a function that does the following:

- Declare and instantiate a ifstream called fin, which opens a text file “data.txt”
- Store all the numbers in the file in an array (assume max of 100 numbers)
- Print out the average
- Close the object fin

You may assume the `data.txt` file looks like this:

35

67

... // random numbers

32

Implementation

Write a function `countLowercase` that does the following:

- Takes in a string from the user
- returns a count of all the lowercase numbers in the string

Using `countLowercase`, write a new function `compareTwoStrings` that:

- Prints which string has more lowercase letters between the two strings
- `compareTwoStrings("Hello", "World") == "Hello has more lowercases than "World"`
- `compareTwoStrings("CSCI", "MATH") == "CSCI and MATH have the same number of lowercase."`

Using `countLowercase`, write a function that:

- Extracts the lowercase letters in a string
- Returns the strictly lowercase letters in order of that string

Implementation

Write a function `longAndShort` that does the following:

- Takes in an array of strings
- Returns a concatenation of the longest, followed by an underscore, followed by the shortest length strings in the array.
- You may assume that there is only 1 longest, and 1 shortest element in the array

Example:

- `string strs[] = {"hey", "hi", "happy"};`
- `longAndShort(strs) == "happy_hi"`